

Probability HW3

Problem #1

$$X = N_H - N_T$$

a) for a coin that is tossed n times, $X = \{-n, -n+2, -n+4, \dots, n-4, n-2, n\}$

b) for $n=3$, $X = \{-3, -1, 1, 3\}$

$$P(X=-3) = \{TTT\} = \frac{1}{8}$$

$$P(X=-1) = \{TTH, HTT, THT\} = \frac{3}{8}$$

$$P(X=1) = \{HHT, THH, HTH\} = \frac{3}{8}$$

$$P(X=3) = \{HHH\} = \frac{1}{8}$$

X	-3	-1	1	3
P	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

c)

$$P(X=x) = \begin{cases} \frac{3}{8} & x = -1, 1 \\ \frac{1}{8} & x = -3, 3 \end{cases}$$

d)

$$F(x) = \begin{cases} 0, & x < -3 \\ \frac{1}{8}, & -3 \leq x < -1 \\ \frac{1}{2}, & -1 \leq x < 1 \\ \frac{7}{8}, & 1 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$

Problem #2

$$a) P(X_1=x) = \begin{cases} \frac{3}{16}, & x=0 \\ \frac{2}{16}, & x=1 \\ \frac{5}{16}, & x=2 \\ \frac{6}{16}, & x=3 \end{cases}$$

$$P(X_2=x) = \begin{cases} \frac{1}{2}, & x=1 \\ \frac{1}{2}, & x=2 \end{cases}$$

$$b) E[X_1] = 0 \cdot \frac{3}{16} + 1 \cdot \frac{2}{16} + 2 \cdot \frac{5}{16} + 3 \cdot \frac{6}{16}$$

$$= 0 + \frac{2}{16} + \frac{10}{16} + \frac{18}{16} = \frac{30}{16} = \frac{15}{8}$$

$$E[X_2] = 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} = \frac{3}{2}$$

Problem #3

$N_1 = \{1, 2, 3\}$ $N_2 = \{1, 2, 3\}$ Let D = Defect, N = no problem

$$P(N_1=1, N_2=1) = \frac{3}{5} \cdot \frac{1}{2} = \frac{3}{10}$$

$$P(N_1=1, N_2=2) = \frac{3}{5} \cdot \frac{1}{2} \cdot \frac{2}{3} = \frac{2}{10}$$

$$P(N_1=1, N_2=3) = \frac{3}{5} \cdot \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{10}$$

$$P(N_1=2, N_2=1) = \frac{2}{5} \cdot \frac{3}{4} \cdot \frac{2}{3} = \frac{2}{10}$$

$$P(N_1=2, N_2=2) = \frac{2}{5} \cdot \frac{3}{4} \cdot \frac{1}{3} = \frac{1}{10}$$

$$P(N_1=2, N_2=3) = 0$$

$$P(N_1=3, N_2=1) = \frac{2}{5} \cdot \frac{1}{4} = \frac{1}{10}$$

$$P(N_1=3, N_2=2) = 0$$

$$P(N_1=3, N_2=3) = 0$$

NDND

NNDD

$N_1 \backslash N_2$	1	2	3
1	$\frac{3}{10}$	$\frac{2}{10}$	$\frac{1}{10}$
2	$\frac{2}{10}$	$\frac{1}{10}$	0
3	$\frac{1}{10}$	0	0

Problem #4

Define $I_k = \begin{cases} 1, & \text{if } k\text{th child is taller than all children before,} \\ 0, & \text{otherwise} \end{cases}$ $k=1, 2, \dots, n$

$X = I_1 + I_2 + I_3 + \dots + I_n$, for k children, each child is equally as likely to be the tallest child

$$P(I_k=1) = \frac{1}{k}$$

$$E[X] = E[I_1] + E[I_2] + \dots + E[I_n] = \sum_{k=1}^n \frac{1}{k}$$

Problem #5

$X_i = \begin{cases} 1, & \text{if neither couple } i \text{ dies} \\ 0, & \text{otherwise} \end{cases}$

$$X = X_1 + X_2 + \dots + X_{100}$$

$$P(X_i = 1) = \frac{150}{200} \cdot \frac{149}{199} = \frac{447}{796}$$

$$E[X] = \sum_{i=1}^{100} E[X_i] = 100 \cdot \frac{447}{796} \hat{=} 56.156$$